

Discussion 6 Solution

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October 2025

Problem 1: Rank and Solution Analysis

Let $\mathbf{A}$ be a 3×3 matrix and $\mathbf{b} \in \mathbb{R}^3$. Consider the linear system $\mathbf{Ax} = \mathbf{b}$. Given

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 6 \\ 12 \\ 3 \end{bmatrix}$$

Analyze the following questions based on the given system:

1. Compute the rank of $\mathbf{A}$.

- **Step 1:** Construct the augmented matrix.

$$[\mathbf{A} \mid \mathbf{b}] = \begin{bmatrix} 1 & 2 & 3 & 6 \\ 2 & 4 & 6 & 12 \\ 1 & 1 & 1 & 3 \end{bmatrix}$$

- **Step 2:** Obtain the RREF augmented matrix.

$$\begin{aligned} \begin{bmatrix} 1 & 2 & 3 & 6 \\ 2 & 4 & 6 & 12 \\ 1 & 1 & 1 & 3 \end{bmatrix} &\sim \begin{bmatrix} 1 & 2 & 3 & 6 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 3 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & -2 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

- **Step 3:** Count the number of columns with pivots in the coefficient matrix.
Column vectors with pivots:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

Hence, the rank of coefficient matrix: 2.

2. Compute the rank of the augmented matrix $[\mathbf{A}|\mathbf{b}]$.

Count the number of columns with pivots in the augmented matrix.

Column vectors with pivots:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

Hence, the rank of augmented matrix: 2.

3. Compare the ranks of $\mathbf{A}$ and $[\mathbf{A}|\mathbf{b}]$. What does this result imply?

From the previous steps, we found:

- The rank of the coefficient matrix $\mathbf{A}$ is 2.
- The rank of the augmented matrix $[\mathbf{A}|\mathbf{b}]$ is also 2.

This result implies that the constant vector $\mathbf{b}$ lies within the column space of $\mathbf{A}$ —that is, it can be expressed as a linear combination of the linearly independent columns of $\mathbf{A}$. It indicates that we can find $\mathbf{x} = (x_1, x_2, x_3)^\top$ such that

$$\mathbf{Ax} = x_1 \times \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + x_2 \times \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix} + x_3 \times \begin{pmatrix} 3 \\ 6 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 12 \\ 3 \end{pmatrix} = \mathbf{b}.$$

Therefore, the key insight from this example is:

If the linearly independent column vectors of the coefficient matrix can generate the constant vector, then the system has at least one solution. This is reflected in the case where $\text{rank}([\mathbf{A}|\mathbf{b}]) = \text{rank}(\mathbf{A})$.

In our case, since both ranks are equal, the system is solvable. Moreover, because the rank is less than the number of unknowns ($2 < 3$), the system has infinitely many solutions.

Problem 2: Rank and Solution Analysis

Let $\mathbf{A}$ be a 3×4 matrix and $\mathbf{b} \in \mathbb{R}^3$. Consider the linear system $\mathbf{Ax} = \mathbf{b}$. Given

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 1 & 1 & 1 & 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 1 \\ 7 \end{bmatrix}$$

Answer the following:

1. Compute the rank of $\mathbf{A}$.

- **Step 1:** Construct the augmented matrix.

$$[\mathbf{A} \mid \mathbf{b}] = \begin{bmatrix} 1 & 0 & 2 & 1 & 4 \\ 0 & 1 & -1 & 2 & 1 \\ 1 & 1 & 1 & 3 & 7 \end{bmatrix}$$

- **Step 2:** Obtain the RREF augmented matrix.

$$\begin{aligned} \begin{bmatrix} 1 & 0 & 2 & 1 & 4 \\ 0 & 1 & -1 & 2 & 1 \\ 1 & 1 & 1 & 3 & 7 \end{bmatrix} &\sim \begin{bmatrix} 1 & 0 & 2 & 1 & 4 \\ 0 & 1 & -1 & 2 & 1 \\ 0 & 1 & -1 & 2 & 3 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 0 & 2 & 1 & 4 \\ 0 & 1 & -1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

- **Step 3:** Count the number of columns with pivots in the coefficient matrix.
Column vectors with pivots:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

Hence, the rank of coefficient matrix: 2.

2. Compute the rank of the augmented matrix $[\mathbf{A}|\mathbf{b}]$.

Count the number of pivots in the augmented matrix.

Column vectors with pivots:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix}$$

Hence, the rank of augmented matrix: 3.

3. Compare the ranks of $\mathbf{A}$ and $[\mathbf{A}|\mathbf{b}]$. What does this result imply?

From the RREF of the augmented matrix, we observe:

- $\text{rank}(\mathbf{A}) = 2$
- $\text{rank}([\mathbf{A}|\mathbf{b}]) = 3$ due to a pivot in the last column

This implies that the constant vector $\mathbf{b}$ cannot be generated by a linear combination of the linearly independent columns of $\mathbf{A}$. In other words, $\mathbf{b}$ lies outside the column space of $\mathbf{A}$. Therefore, the system $\mathbf{A}\mathbf{x} = \mathbf{b}$ has no solution. This illustrates the principle that if the linearly independent vectors of the coefficient matrix cannot generate the constant vector, then the system is inconsistent.

Problem 3: Transpose and Inverse

Prove that the inverse of $\mathbf{A}$ is $\mathbf{A}^\top$ for any value of θ if $\mathbf{A}$ is

$$\mathbf{A} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}.$$

Hint: $\cos^2(\theta) + \sin^2(\theta) = 1$ for any θ .

$$\begin{aligned} \mathbf{A}^{-1} &= \frac{1}{\cos^2(\theta) + \sin^2(\theta)} \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \\ &= \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \\ &= \mathbf{A}^\top \end{aligned}$$

Problem 4: Determinant

Compute determinant for $\mathbf{A}$ for each case:

$$1. \mathbf{A} = \begin{bmatrix} a & 0 \\ b & c \end{bmatrix}$$

$$\begin{aligned} |\mathbf{A}| &= ac - b \cdot 0 \\ &= ac \end{aligned}$$

$$2. \mathbf{A} = \begin{bmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{bmatrix}$$

$$\begin{aligned} |\mathbf{A}| &= (-1)^{3+3} \cdot f \cdot \begin{vmatrix} a & 0 \\ b & c \end{vmatrix} + (-1)^{2+3} \cdot 0 \cdot \begin{vmatrix} a & 0 \\ d & e \end{vmatrix} + (-1)^{1+3} \cdot 0 \cdot \begin{vmatrix} b & c \\ d & e \end{vmatrix} \\ &= f \cdot ac + 0 + 0 \\ &= acf \end{aligned}$$

$$3. \mathbf{A} = \begin{bmatrix} a & 0 & 0 & 0 \\ b & c & 0 & 0 \\ d & e & f & 0 \\ g & h & i & j \end{bmatrix}$$

$$\begin{aligned} |\mathbf{A}| &= (-1)^{4+4} \cdot j \cdot \begin{vmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{vmatrix} + (-1)^{3+4} \cdot 0 \cdot \begin{vmatrix} a & 0 & 0 \\ b & c & 0 \\ g & h & i \end{vmatrix} + (-1)^{2+4} \cdot 0 \cdot \begin{vmatrix} a & 0 & 0 \\ d & e & f \\ g & h & i \end{vmatrix} + (-1)^{1+4} \cdot 0 \cdot \begin{vmatrix} b & c & 0 \\ d & e & f \\ g & h & i \end{vmatrix} \\ &= j \cdot acf + 0 + 0 + 0 \\ &= acfj \end{aligned}$$

4. Generalize the determinant of $n \times n$ lower triangular matrix $\mathbf{A}$.

$$|\mathbf{A}| = \prod_{i=1}^n a_{ii}$$

where a_{ii} is the i th diagonal entry of $\mathbf{A}$.